

Supplementary Material

Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Audited Spatial Scoring

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Table S1: Full per-perturbation score-drop statistics. Nuisance rows use absolute drop; structural rows use signed drop from the identity score. Confidence intervals use reference-level bootstrap resampling.

Perturbation	N	Mean	Median	SD	95% CI	P05-P95	Invalid	ρ_s	Monotonic
Translation	800	0.000	0.000	0.000	[0.000, 0.000]	[0.000, 0.000]	0.000	–	1.000
Rotation	520	5.395	4.890	2.335	[5.079, 5.747]	[3.045, 9.024]	0.133	-0.521	0.647
Scale up	408	13.653	12.775	4.556	[12.964, 14.420]	[8.030, 22.745]	0.490	0.812	0.271
Scale down	800	16.356	15.412	5.225	[15.670, 17.074]	[10.051, 26.419]	0.000	-0.662	0.500
Compound transform	276	8.143	7.298	3.468	[7.621, 8.717]	[4.295, 15.464]	0.540	0.397	0.466
Terminal deletion	800	19.757	11.113	20.424	[17.387, 22.087]	[0.403, 62.702]	0.000	-0.993	0.992
Added direction fragment	800	11.398	6.935	12.736	[9.934, 12.929]	[0.277, 41.357]	0.000	-0.988	0.987
Local fragment shift	800	4.310	2.537	4.220	[3.886, 4.772]	[0.547, 12.677]	0.000	-0.985	0.985
Direction dilation	800	24.034	17.866	18.673	[22.180, 25.955]	[3.336, 62.536]	0.000	-1.000	1.000
Direction erosion	800	22.225	15.425	18.048	[20.269, 24.344]	[3.550, 60.381]	0.000	-0.992	0.997
Keypoint shift	800	7.468	4.570	6.971	[7.294, 7.641]	[0.000, 18.232]	0.000	-1.000	1.000

Table S2: Paired alignment comparisons by perturbation family. Positive benefit differences favour the current constrained alignment.

Comparison	Perturbation	Refs	Ours	Comparator	Benefit 95% CI	p	RBC
Ours–none	compound transform	156	8.143	43.947	[34.353, 37.172]	< 0.001	1.000
Ours–none	terminal deletion	200	19.757	2.135	[15.447, 19.802]	< 0.001	1.000
Ours–none	direction dilation	200	24.034	4.624	[17.685, 21.234]	< 0.001	1.000
Ours–none	direction erosion	200	22.225	4.641	[15.636, 19.565]	< 0.001	1.000
Ours–none	added fragment	200	11.398	0.923	[9.021, 11.985]	< 0.001	1.000
Ours–none	rotation	180	5.395	15.927	[10.176, 10.871]	< 0.001	1.000
Ours–none	scale down	200	16.356	48.355	[31.504, 32.517]	< 0.001	1.000
Ours–none	scale up	170	13.653	35.869	[20.706, 23.708]	< 0.001	0.988
Ours–none	translation	200	0.000	43.034	[42.054, 44.005]	< 0.001	1.000
Ours–none	keypoint shift	200	7.468	7.468	[0.000, 0.000]	1.000	0.000
Ours–none	local fragment shift	200	4.310	2.270	[1.684, 2.427]	< 0.001	1.000
Ours–wide	compound transform	156	8.143	22.951	[14.336, 15.312]	< 0.001	1.000
Ours–wide	terminal deletion	200	19.757	20.231	[-0.639, -0.317]	< 0.001	-0.696
Ours–wide	direction dilation	200	24.034	24.605	[-0.751, -0.410]	< 0.001	-0.623
Ours–wide	direction erosion	200	22.225	22.730	[-0.682, -0.341]	< 0.001	-0.577
Ours–wide	added fragment	200	11.398	11.560	[-0.280, -0.054]	0.005	-0.407
Ours–wide	rotation	180	5.395	12.019	[6.344, 6.908]	< 0.001	1.000
Ours–wide	scale down	200	16.356	16.861	[0.439, 0.571]	< 0.001	0.919
Ours–wide	scale up	170	13.653	14.107	[0.377, 0.533]	< 0.001	0.841
Ours–wide	translation	200	0.000	0.000	[0.000, 0.000]	1.000	0.000
Ours–wide	keypoint shift	200	7.468	7.468	[0.000, 0.000]	1.000	0.000
Ours–wide	local fragment shift	200	4.310	4.302	[-0.006, 0.024]	0.263	0.444